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MILITARY SPECIFICATION

NIGHT VISION SIGHT, CREW SERVED WEAPON AN/TVS-5

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the Night Vision Sight, Crew Served Weapon AN/TVS-5.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards and handbooks. The following specifications, standards and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2.k).

SPECIFICATIONS

FEDERAL

H-B-1654 - Brush, Dusting, Lens and Photographic Negatives

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: HQ, USA Communications-Electronics Command, ATTN: AMSEL-ED-TM, Fort Monmouth, NJ 07703-5023 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A

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FEDERAL

NNN-P-40	-	Paper, Lens

MILITARY

MIL-P-11268	-	Parts, Materials, and Processes used in Electronic Equipment
MIL-M-13231	-	Marking of Electronic Items
MIL-B-49030/6	-	Battery, Dry, BA-3058/U
MIL-I-49040	-	Image Intensifier Assembly, 25 Millimeter, Microchannel Inverter, MX-9644/UV
MIL-L-49367	-	Lens Assembly, Objective, 15mm AN/TVS-5
MIL-L-49368	-	Lens Assembly, Eyepiece, AN/PVS-4 and AN/TVS-5
MIL-B-49430/4	-	Battery, Non-Rechargeable, Lithium Sulfur Dioxide, BA-5567/U

STANDARDS

MILITARY

MIL-STD-150	-	Photographic Lenses
MIL-STD-171	-	Finishing of Metal and Wood Surfaces
MIL-STD-202	-	Test Methods for Electronic and Electrical Component Parts
MIL-STD-454	-	Standard General Requirements for Electronic Equipment
MIL-STD-810	-	Environmental Test Methods
DOD-STD-2000A/4	-	General Purpose Soldering Requirements for Electrical and Electronic Equipment

(Unless otherwise indicated, copies of federal and military specifications, standards and handbooks are available from the Standardization Document Order Desk, 700 Robbins Avenue, Bldg #4, Section D, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings and publications. The following other Government documents, drawings and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation (see 6.2.k).

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DRAWINGS

USA COMMUNICATIONS-ELECTRONICS COMMAND

SM-D-850081	-	Daylight Cover
SM-D-850101	-	Night Vision Sight, Individual Served Weapon AN/TVS-5
SM-D-850111	-	Housing, Battery
SM-D-850121	-	Eyeguard Assembly
SM-D-850181	-	Right Angle Relay
SM-D-850201	-	Case, Carrying and Storage
SM-D-850202	-	Bag, Carrying
A3009873	-	BA-3058/U ("AA") Battery Adapter

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Description. The Night Vision Sight, Crew Served Weapon AN/TVS-5, herein referred to as the sight, is intended for use at night in aimed firing of crew served weapons.

3.2 Qualification. Unless otherwise specified by the contract or purchase order, Sights furnished under this specification shall be products approved as qualified products (see 3.9.5 and 6.4).

3.2.1 Initial production testing. When specified in the contract (6.2.j), the contractor shall furnish IPT sights in accordance with 4.4.

3.3 Construction. The sight shall be constructed in accordance with SM-D-850101 and as specified herein. Unless otherwise specified, finishing and treatment of metal surfaces shall conform to MIL-STD-171.

3.4 Material. Material shall be as specified herein and as shown on the applicable drawings. Material not specified shall be selected by the contractor, shall be subject to all provisions of this specification, and shall conform to MIL-P-11268.

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3.5 Major components. The following components shall be constructed in accordance with applicable drawings, specifications, and as specified herein.

3.5.1 Objective lens assembly. The objective lens assembly shall conform to MIL-L-49367.

3.5.2 EyePiece lens assembly. The eyepiece lens assembly shall conform to MIL-L-49368.

3.5.3 Eyeguard assembly. The eyeguard assembly (referred to herein as the eyeguard) shall conform to SM-D-850121.

3.5.4 Right angle relay. The right angle relay shall conform to SM-D-850181. The right angle relay-eyepiece lens assembly is referred to herein as the right angle eyepiece.

3.5.5 BA-3058/U ("AA") Battery Adapter. The "AA" Battery Adapter shall conform to A3009873.

3.5.6 Image intensifier assembly. The image intensifier assembly shall conform to MIL-I-49040.

3.5.7 Battery housing. The battery housing shall conform to SM-D-850111 and as specified herein.

3.5.8 Daylight cover. The daylight cover shall conform to SM-D-850081.

3.5.9 Brush. The brush shall conform to H-B-1654.

3.5.10. Lens paper. The lens paper shall conform to NNN-P-40.

3.5.11 Case. The case, carrying and storage, shall conform to SM-D-850201.

3.5.12 Bag. The bag, carrying, shall conform to SM-D-850202.

3.6 Operation and environment.

3.6.1 Eyeguard operation. With the eyeguard vertical and at -48°C, the eyeguard shall open (depress) when a 2000, ± 10 gram weight is placed axially on the eye flange and shall close (expand) when the weight is reduced to 100, ± 10 grams. At +23°C, the eyeguard shall open (depress) when a 1600, ± 10 gram weight is placed axially on the eye flange and shall close (expand) when the weight is reduced to 100, ± 10 grams. At +23°C and with the eyeguard open (depressed) the minimum open position of the 2 leaves shall be 1.875 inches.

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3.6.2 Right angle eyepiece. When fitted to an eyepiece lens assembly as specified herein, the right angle relay-eyepiece lens combination shall meet the following requirements.

3.6.2.1 Equivalent focal length. The equivalent focal length of the right angle eyepiece at zero diopter shall be 26.5, ± 0.10mm.

3.6.2.2 Flange focal distance. The flange focal distance of the right angle eyepiece shall be 4.753 (+0.207, -0.435)mm.

3.6.2.3 Linear distortion. The linear distortion of the right angle eyepiece at the edge of the 25 mm format shall be between zero and 5 percent pincushion distortion.

3.6.2.4 Resolution. When attached to a sight (see 3.6.3), the right angle eyepiece shall not degrade the resolution of the sight (see 3.6.4) by more than 2 line pairs/mm referenced to sight resolution.

3.6.2.5 Exit pupil. The exit pupil diameter of the right angle eyepiece shall be 5, ± 0.2mm at an eye relief of 28, ± 0.5mm for zero diopter setting.

3.6.2.6 Field flatness. The field flatness of the right angle eyepiece shall deviate no more than ± 1/4 diopter over the full 25mm format.

3.6.2.7 Transmission. The transmission of the right angle eyepiece shall not be less than 70 percent over the full eyepiece aperture for the spectral output of the P-20 phosphor.

3.6.2.8 Modulation transfer function (MTF). The MTF of the right angle eyepiece shall be not less than that specified in Figure 1.

3.6.2.9 Environmental performance. The right angle eyepiece, when attached to a sight meeting the requirements of this specification (see 6.2) shall not be damaged (see 3.9.1), shall meet all the environmental requirements of 3.6.10 and 3.6.11, and shall not degrade the performance of the sight as specified herein.

3.6.3 Sight definition. For conformance to the following performance parameters, the sight is defined as sight less case, bag and ancillary equipment.

3.6.4 Resolution. The minimum sight resolution shall be within one element number of the resolution of the image intensifier used in the sight at the time of testing while using a USAF 1951 Resolving Power Test Target.

3.6.5 Field of view. The sight shall have a minimum field of view of 156 milliradians.

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3.6.6 Vibration. The sight shall withstand simple harmonic motion having an amplitude of 0.015 inch (0.030 inch total excursion) with the frequency being varied between 5 and 55 hertz (Hz).

3.6.7 Weapon shock. The sight, mounted and zeroed (see 3.9.2) on the M2 .50 Caliber Machine Gun, shall not shift in zero more than ± 0.50 milliradian or be damaged (see 3.9.1) when subjected to vibrations encountered during machine gun firing.

3.6.8 Weight. The weight of the sight with batteries but without daylight cover shall not exceed 8.5 pounds.

3.6.9 Controls and adjustments. All controls and adjustments shall operate smoothly without binding throughout their complete range of movement at all operating temperatures specified herein.

3.6.9.1 Eyepiece diopter focus. With the diopter adjustment control set at -6 (+1/4, -0.0) to +2 diopter the image of the projected reticle shall be in sharp focus.

3.6.9.2 Range focus. The sight shall provide a minimum focus adjustment range from a near focus of 25 meters to infinity. The dynamic torque to turn the range focus control shall be 8 ± 7 inch-pounds.

3.6.9.3 Tube brightness control. The tube brightness control shall vary the gain of the image intensifier tube from OFF to maximum gain and control the power to the reticle brightness control.

3.6.9.4 Reticle brightness control. Brightness of the reticle pattern shall be continuously variable over the range from OFF to full brightness. The pattern shall be visible and the numbers legible when viewing a solid white target having a brightness of 3.3×10^{-3} footlamberts minimum.

3.6.9.5 Reticle movement. (see 6.2.b) Two external control knobs shall control the position of the reticle pattern viewed by the observer using the sight. One control shall move the reticle pattern in azimuth; another control shall move the reticle pattern in elevation. An adjustment made by turning one control shall not cause the reticle to deviate from a straight line of travel by more than 0.5 milliradian. Adjustment will be apparent to the operator by tactile indexing with audible clicks. Each click of the control knob shall displace the reticle pattern 0.25, ± 0.04 milliradian. Total movement of the reticle pattern shall be a minimum of 2.5° in any direction from the optical axis. The torque required to adjust the reticle position shall be between 0.5 and 4.0 inch-pounds. The reticle pattern shall remain in focus throughout the full range of movement of the range focus control. The reticle pattern shall not exhibit rotation greater than 1° from its proper orientation (see Figure 2).

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3.6.9.6 Battery adapter. The sight shall properly operate with the battery adapter attached and Type BA-3058/U batteries conforming to MIL-B-49030/6 inserted.

3.6.10 High temperature storage and operation. The sight shall not be damaged (see 3.9.1) by storage in any temperature from +23°C to +68°C and shall operate as specified herein at any temperature from +23°C to +52°C.

3.6.11 Low temperature storage and operation. The sight shall not be damaged (see 3.9.1.) by storage in any temperature from +23°C to -51°C and shall operate as specified herein at any temperature from +23°C to -51°C.

3.6.12 Altitude. The sight shall not be damaged (see 3.9.1) when operated at a pressure equivalent to 10,000 feet altitude above sea level.

3.6.13 Temperature shock. The sight shall not be damaged (see 3.9.1) after being subjected to temperature changes between +68°C, and +23°C in a 5 minute period and between +23°C and -51°C in a 5 minute period.

3.6.14 Humidity. The sight shall not be damaged (see 3.9.1) when operated in an atmosphere with a relative humidity of 95 percent, ± 5 percent over a temperature range of +20°C to +52°C and shall operate after storage up to +68°C minimum with 95 percent, ± 5 percent relative humidity.

3.6.15 Immersion. The sight shall not be damaged (see 3.9.1) after being immersed in fresh water to a depth of not less than 3 feet for a period of not less than 30 minutes.

3.6.16 High intensity shock. The sight shall not be damaged (see 3.9.1) by a sequence of shocks applied in each direction along each of 3 mutually perpendicular axes (horizontal, vertical, and optical axes when the reticle pattern is erect). The shocks shall be half sine pulses and shall have a time duration of 4 milliseconds, ± 5 percent. Shock pulses shall have a peak amplitude of 100g's, ± 15g's (see 3.9.3).

3.6.17 Interchangeability. The objective lens assembly and the eyepiece lens assembly shall be physically interchangeable between sight assemblies without degradation of the resolution (see 3.6.4), eyepiece diopter focus (see 3.6.9.1), and range focus (see 3.6.9.2) requirements. These interchanges shall be made without any modification or adjustment of any of the parts or assemblies involved.

3.7 Marking. Unless otherwise specified, markings on the sight and all components shall conform to MIL-M-13231.

3.8 Workmanship. Unless otherwise specified, workmanship of the sight and ancillary equipment shall conform to MIL-STD-454, Requirement 9. Soldering shall conform to the requirements of DOD-STD-2000A/4.

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3.9 Technical interpretations. The following technical interpretations are, when referenced in sections 3, 4, or 5, mandatory for this specification.

3.9.1 Damage. Breakage, loosening, shifting, evidence of corrosion, or failure of any finish, hardware, connection or component; leakage or condensation of moisture within the viewer; any degradation in input or output characteristics; and any failure of the control knobs to perform their function.

3.9.2 Zero. A known and specified relationship between the sight aiming point and the strike of the bullet.

3.9.3 "g". "g" is a unitless measure of the quantity of units of force associated with the earths gravity that a object is subjected to.

3.9.4 Room temperature. Room temperature is defined as +23°C, ± 3°C.

3.9.5 Qualified product. A product which has successfully met all of the requirements of an IPT or First Article witnessed by the CECOM, Concurrent Engineering Directorate or its designated representative. Should the supplier make substantial changes in the process, personnel, location, or equipment used to produce a qualified product, or have not delivered the qualified product to the Federal Government within 12 months of contract award, an IPT shall be performed to the extent necessary to verify continued compliance to specification requirements for all characteristics of the product affected by the changes.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of section 3 and 5. The inspection set forth in this specification shall not become part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract.

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Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.1.2 Components and materials inspection. The contractor is responsible for insuring that components and materials used are manufactured, examined, and tested in accordance with referenced specifications and standards as specified herein.

4.2 Classification of inspection. Inspection shall be classified as follows:

a. Initial production testing inspection (4.4).
b. Quality conformance inspection (4.5).

4.3 Inspection conditions. Unless otherwise specified in the contract or purchase order, inspections shall be performed as specified herein. Tolerances on specified illumination levels shall be plus or minus 10 percent. Neutral density filters used in test equipment shall have transmission characteristics within 10 percent of the nominal filter transmission from 0.35 to 1.0 micrometer. When a collimator is used, equivalent focal length (EFL) of the collimator shall be at least six times the EFL of the objective lens under test. The clear aperture of the collimator shall be six inches or greater. Tolerance on 2854K color temperature shall be ± 200K. References within environmental requirements and tests to "initial measurements", "operate", "pretest", or "performance check during test", shall be defined as the resolution test 4.6.4 unless otherwise specified.

4.4 Initial production testing.

4.4.1 IPT samples. The quantity of initial production testing samples shall be specified in the contract or purchase order and shall be selected from the first production lot. Disposition of IPT samples shall be in accordance with the contract or purchase order (see 6.2.g).

4.4.1.1 Inspection. Each sight shall be inspected in process in accordance with Table I and 3.3, 3.4, 3.7, and 3.8. Presence of one or more defects shall be cause for rejection of that sight.

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TABLE I. Inspections.

Requirements
Performance requirements specified in notes appearing on drawings.
Linear or diametral tolerances of 0.002 inch or less.
Angular dimension having a tolerance of 15 minutes of arc or less.
Machine finishes of 32 micro-inches RMS or finer.
Tolerances of position or form 0.002 inch or less.
Screw thread Class 3 or better.
Load requirements for springs.
Lubrication requirements including dry lubricants on finishes.
Material hardness when specified on drawings.
Soldering shall conform to DOD-STD-2000A/4.

4.4.2 Tests. Following successful completion of the inspections specified in 4.4.1, each IPT sight shall be subjected to all inspections in Table II. Failure of any inspections shall be cause for rejection of that sight. Inspections may be conducted in any order except resolution shall be performed last.

TABLE II. Initial production testing inspection.

Inspection	Requirement paragraph	Test paragraph
Workmanship	3.8	
Eyeguard operation	3.6.1	4.6.1
Right angle eyepiece	3.6.2	4.6.2
Field of view	3.6.5	4.6.5
Vibration	3.6.6	4.6.6
Weapon Shock	3.6.7	4.6.7
Weight	3.6.8	4.6.8
Eyepiece diopter focus	3.6.9.1	4.6.9.1
Range focus	3.6.9.2	4.6.9.2
Tube brightness control	3.6.9.3	4.6.9.3
Reticle brightness control	3.6.9.4	4.6.9.4
Reticle movement	3.6.9.5	4.6.9.5
High temperature storage		4.6.10
and operation	3.6.10	4.6.10
Low temperature storage		4.6.11
and operation	3.6.11	4.6.11
Altitude	3.6.12	4.6.12
Temperature shock	3.6.13	4.6.13
Humidity	3.6.14	4.6.14
Immersion	3.6.15	4.6.15
High intensity shock	3.6.16	4.6.16
Interchangeability	3.6.17	1.6.17
Resolution	3.6.4	4.6.4

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4.5 Quality conformance inspection.

4.5.1 Group A inspection. Following completion of the inspections of 4.4.1.1 and Table I, each sight that has passed the inspections shall be inspected per Table III. Failure of any sight to pass all the tests of Table III shall be cause for rejection of that sight.

TABLE III. Group A inspections.

Inspection	Requirement paragraph	Test paragraph
Workmanship	3.8	3.8
Tube brightness control	3.6.9.3	4.6.9.3
Reticle brightness control	3.6.9.4	4.6.9.4

4.5.2 Group B inspection. Sights which have passed the inspections specified in 4.5.1 shall be formed into lots by the contractor. Sample sights shall be selected and subjected to the inspection of Table IV. Sampling shall be in accordance with the contract or purchase order (see 6.2.d). Inspections shall be performed in the order listed.

TABLE IV. Group B inspection.

Inspection	Requirement paragraph	Test paragraph
Right angle eyepiece*	3.6.2	4.6.2
Eyepiece diopter focus	3.6.9.1	4.6.9.1
Resolution	3.6.4	4.6.4
Range focus	3.6.9.2	4.6.9.2
Reticle movement	3.6.9.5	4.6.9.5

* If purchased

4.5.2 Group C inspection. Group C inspections shall be conducted on sights selected from lots which have passed the inspections in 4.5.1. The sample(s) shall be subjected to the inspections specified in Table V. Samples shall be selected in accordance with the contract or purchase order (see 6.2.d). Tests may be performed in any order except that in Subgroup 1, Immersion shall follow Vibration, and in Subgroup 2, Humidity shall be last.

4.5.3.1 Weapon shock testing. Subgroup 3, Weapon shock testing shall be performed on a sample of two (2) sights selected from each lot of sights.

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TABLE V. Group C inspection.

Inspection	Requirement paragraph	Test paragraph
Subgroup 1		
High temperature storage and operation	3.6.10	4.6.10
Low temperature storage and operation	3.6.11	4.6.11
Vibration	3.6.6	4.6.6
Immersion	3.6.15	4.6.15
Subgroup 2		
High intensity shock	3.6.16	4.6.16
Temperature shock	3.6.13	4.6.13
Humidity	3.6.14	4.6.14
Subgroup 3		
Weapon shock	3.6.7	4.6.7

4.5.4 Group C failures. Actions required relative to Group C failure shall be as specified in the contract or purchase order (see 6.2.f).

4.5.5 Disposition of Group C inspection units. Any sights which have been subjected to Group C testing shall be subjected (after refurbishment, if necessary) to the test of Tables III and IV prior to acceptance. Failure of a sight of any test of Tables III and IV shall be cause to reject that sight.

4.6 Test methods.

4.6.1 Eyeguard operation.

4.6.1.1 Low temperature compression/expansion. Place the eyeguard in an environmental chamber at -48°C, ± 3°C for 2 hours. Without removing the eyeguard form the chamber, place a 2000, ± 10 gram weight axially on the eye flange. Reduce the weight to 100, ± 10 grams. Failure of the eyeguard to (depress) within 5 seconds after adding the 2000 gram weight and to close after reducing the weight to 100 grams, or failure to meet the requirements of 3.6.1, shall constitute failure of this test.

4.6.1.2 Room temperature compression/expansion. At +23°C, repeat the inspection of 4.6.1.1 using a 1600, ± 10 gram weight. Failure of the eyeguard to open (depress) within 3 seconds after adding the 1600 gram weight and to close after reducing the weight to 100, ± 10 grams, or failure to meet the requirements of 3.6.1 shall constitute failure of this test.

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4.6.1.3 Leaf separation. During inspection of 4.6.1.1 and 4.6.1.2, measure the leaf separation. Failure of the leaf separation to be 1.875 inches or greater, or failure to meet the requirements of 3.6.1, shall constitute failure of this test.

4.6.2. Right angle eyepiece.

4.6.2.1 Equivalent focal length. The equivalent focal length of the right angle relay shall be measured in accordance with MIL-STD-150. Failure to comply with the requirements of 3.6.2.1 shall constitute failure of this test.

4.6.2.2 Flange focal distance. The flange focal distance of the right angle relay shall be measured in accordance with MIL-STD-150. Failure to comply with the requirements of 3.6.2.2 shall constitute failure of this test.

4.6.2.3 Linear distortion. The linear distortion of the right angle relay shall be measured using procedures outlined in MIL-STD-150. In order to use these procedures, the right angle relay shall be run backwards (i.e. collimated light enters the eyelens) with a 5mm minimum exit pupil at a 28, ± 0.5mm eye relief distance between the collimator and the right angle relay. Failure to meet requirement of 3.6.2.3 shall constitute failure of this test.

4.6.2.4 Resolution. With the right angle eyepiece attached to a sight, the resolution shall be measured using the procedure of 4.6.4. Measurement of resolution less than 2 lines pairs/mm below the sight resolution, or failure to meet the requirements of 3.6.2.4, shall constitute failure of this test.

4.6.2.5 Exit pupil. Brightly illuminate a cross target at the infinity focal plane of a collimator. Position an opaque mask on the eyepiece at the 29, ± 0.5mm eye relief distance. The mask shall have two pinholes (1mm diameter) 4mm apart (center to center), centered on the optical axis, and aligned in the azimuth direction. Place the eyepiece on a nodal slide so that the mask is between the collimator and the eyepiece. Rotate the eyepiece in azimuth 27°, ± 1° in each direction and locate the image of the cross target using a traveling microscope of 20X or greater. Failure to observe the cross target through each of the pinholes at either +27° or -27° rotation from center or failure to meet the requirements of 3.6.2.5, shall constitute failure of this test.

4.6.2.6 Field flatness. Field flatness of the right angle relay shall be measured in accordance with MIL-STD-150. In order to use this procedure, the right angle relay shall be run backwards (i.e. collimated light enters the eyelens with a 5mm minimum exit pupil at a 28, ± 0.5mm eye relief distance between the collimator and the right angle relay). Failure to comply with requirements of 3.6.2.6 shall constitute failure of this test.

4.6.2.7 Transmission. Transmission of the right angle relay shall be measured in accordance with MIL-STD-150. Failure to comply with the requirements of 3.6.2.7 shall constitute failure of this test.

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4.6.2.8 Modulation transfer function. A sine wave pattern generator with an S-11 photomultiplier tube shall be used. A green filter (Written No. 58 or equivalent) shall simulate P-20 illumination in the setup. If a microscope objective is used to relay the image of the lens, the microscope objective numerical aperture (NA) must be greater than or equal to 0.35 with a minimum magnification of 10X. The pattern-collimator combination must allow the lens to be tested either continuously or at a minimum of 8 distinct, evenly distributed points between 0-40 lp/mm, including 0 lp/mm, referenced to the plane of the right angle relay.

4.6.2.8.1 MTF on-axis, centered pupil. Place a 5mm (minimum) stop at the 28, ± 0.5mm eye relief distance of the right angle relay centered with respect to the right angle relay aperture. Measure MTF for an on-axis position. Failure to meet the requirements of 3.6.2.8 shall constitute failure of this test.

4.6.2.8.2 MTF on-axis, off-centered pupil. Place a 5mm (minimum) stop at the 28, ± 0.5mm eye relief distance of the right angle relay. Off-center this stop 2mm perpendicular to the optical axis and measure on-axis MTF. Off-center this stop 5mm perpendicular to the optical axis and again measure on-axis MTF. Failure to comply with the requirements of 3.6.2.8 shall constitute failure of this test.

4.6.2.8.3 MTF off-axis, centered pupil. Place a 5mm stop at the 28, ± 0.5mm eye relief distance of the right angle relay centered with respect to the right angle relay aperture. Rotate the lens and analyzer to the 2/3 field height position and measure tangential MTF. Failure to comply with the requirements of 3.6.2.8 shall constitute failure of this test.

4.6.2.9 Right angle relay environmental tests. With the right angle eyepiece attached, the sight shall be subjected to the tests of 4.6.10 and 4.6.11. Any damage, failure to operate, or failure to pass inspection, or meet the requirements of 3.6.2.9 shall constitute failure of this test.

4.6.3 Test conditions. Unless otherwise specified, the following test conditions shall apply.

a. Tolerances on specified radiation levels shall be ± 10 percent.

b. Meters used to monitor lamp current or voltage shall be accurate to ± 0.25 percent.

c. Tests shall be performed at room temperature (see 3.9.4) except where otherwise specified.

d. Test chambers used for environmental temperature tests shall maintain temperatures within ± 2.0C.

e. Unless otherwise specified, tests shall be performed using hard source power supplies with a tolerance of ± 0.05 Vdc.

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4.6.4 Resolution. Place a 100 percent contrast USAF 1951 Resolving Power Test Target (black on a clear background) in the focal plane of a collimator having a known focal length. Back illuminate the target with a 2854K white light of such intensity that the brightness of the target through the collimator is 3.3 x 10^{-3} footlamberts. Place the sight with the range focus set for infinity looking into the collimator. Determine the smallest pattern which can be resolved. Failure to meet the requirements of 3.6.4 shall constitute failure of this test.

4.6.5 Field of view. Place two dark vertical lines on a wall or on a pattern in a collimator with 156 milliradian separation with respect to the sight. With the sight stationary, observe through the eyepiece. Inability to see both lines at the edge of the field of view at one time, or failure to meet the requirements of 3.6.5, shall constitute failure of this test.

4.6.6 Vibration. The sight shall be subjected to Method 201A of MIL-STD-202. The test after vibration shall be resolution. Operation during vibration shall not be required. The induced vibratory force shall be applied in each of 3 mutually perpendicular planes one of which is perpendicular to the optical axis. The duration of vibration shall be 5 minutes in each plane. Damage to the sight, failure to pass the resolution test, or failure to meet the requirements of 3.6.6, shall constitute failure of this test.

4.6.7 Weapon firing. This test may be conducted during daylight hours with a daylight cover. The sight shall be mounted on an M2 .50 caliber machine gun by means of a weapon adapter bracket (see 6.2.h) and shall be sighted on a target. The sight shall be zeroed in a manner selected by the Government and all hardware and connectors shall be checked to insure that no parts are loose. No adjustments, tightening of parts or repairs of any kind shall be permitted after the initial zeroing is completed and test firing cycles have begun.

4.6.7.1 Quality conformance weapon firing. The weapon shall be fired in 4 to 10 round bursts, until 100 rounds have been fired. During the firing, evidence of loss of power to image intensifier shall constitute failure of this test. After 100 rounds have been fired, the sight shall be inspected for damage and the position of the reticle in the sight with the relative positioning prior to firing the 100 rounds. Any evidence of damage or loss of zero exceeding 0.5 milliradian, or failure to meet the requirements of 3.6.7, shall constitute failure of this test.

4.6.7.2 IPT weapon firing. The firing and examination of 4.6.7.1 shall be repeated sequentially until 500 rounds have been fired. Failure to conform to the acceptance criteria of 4.6.7.1, or evidence of loss of zero exceeding 0.5 milliradian when compared to the relative position prior to firing the 500 rounds, shall constitute failure of this test.

4.6.8 Weight. Weigh the sight on a scale which can be read to within ± 0.1 pound with an accuracy within ± 1.0 percent. Failure to meet the requirement of 3.6.8 shall constitute failure of this test.

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4.6.9 Controls and adjustments.

4.6.9.1 Eyepiece diopter focus. With the system operating, observe the reticle pattern through the eyepiece with a diopter scope of 3 power or greater set at -5.75 diopter and +2 diopter. Failure to obtain sharp focus at these two diopter settings, or failure to meet the requirements of 3.6.9.1, shall constitute failure of this test.

4.6.9.2 Range focus.

a. For the infinity focus requirement, place a point source of light in the focal plane of a collimator. Observe the point source with the sight. Adjust the range focus in and out, expanding the image size on both sides of best focus. Inability to increase the size of the image on both sides of best focus, or failure to meet the requirements of 3.6.9.2 shall constitute failure of this test.

b. For the minimum focus requirement, place a point source of light at a distance of 25.0, +0.5 meters in front of the objective. Adjust the range focus and observe that the point source can be brought into sharp focus. Inability to bring the 25 meter target into sharp focus shall constitute failure of this test.

c. Measure the dynamic torque required to rotate the range focus. Failure to comply with the requirements of 3.6.9.2 shall constitute failure of this test.

4.6.9.3 Tube brightness control. Attach the sight to a test bench so that a collimator target can be projected directly into the objective assembly. Back illuminate the target located at the focal plane of the collimator at 1 x 10^-5 footcandles. Turn the Tube Brightness control clockwise past the detent position. Open the eyeguard, look into the eyepiece, and adjust the Tube Brightness for Maximum brightness. Failure of the tube brightness to vary as the control is turned, or to meet the requirements of 3.6.9.3 shall constitute failure of this test.

4.6.9.4 Reticle brightness control. With the sight, observe a solid white target of brightness 3.3 x 10^-3 footlamberts in a collimator. Failure to clearly distinguish the reticle pattern, or to meet the requirements of 3.6.9.4 shall constitute failure of this test.

4.6.9.5 Reticle movement. (see 6.2.b) With the same test setup as 4.6.9.3 and with a test target per Figure 3, turn on the reticle brightness control and adjust for optimum illumination of the reticle pattern. Adjust the range focus and diopter ring for a sharp image of the reticle pattern.

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4.6.9.5.1 Adjustment accuracy. With the reticle center on the crosshairs of the target, turn the reticle adjustment 20 clicks to the right and observe the position of the reticle center relative to the two lines A and B. Repeat measurement, moving the reticle 20 clicks to the left of the target center. Repeat these measurements using the up and down reticle adjustment. Failure of the center of the reticle pattern to fall between the two lines A and B for any of the four observations, or failure to meet the requirements of 3.6.9.5, shall constitute failure of this test.

4.6.9.5.2 Adjustment range. With the reticle pattern at its center of movement, aline the test target center crosshairs with the reticle. Turn the azimuth adjustment until the center of the reticle reaches one extreme of the 5 circle and then back to the opposite side of the circle. Return the reticle to the center of the circle. Repeat this procedure using the elevation adjustment. Failure of the reticle to traverse the 5° circle, to travel in a straight line, evidence of binding or slipping of the reticle, failure of the reticle to move continuously with adjustment control rotation, or failure to meet the requirements of 3.6.9.5, shall constitute failure of this test.

4.6.9.5.3 Torque. Using a torque wrench and adapter, rotate the elevation adjustment, first clockwise and then counterclockwise, and observe the readings. Repeat this procedure for the azimuth adjustment. Failure of any of the four torque readings to fall between 0.5 and 4 inch-pounds or failure to hear the audible click or feel the tactile indexing, or failure to meet the requirements of 3.6.9.5 shall constitute failure of this test.

4.6.9.5.4 Alignment. Place a target (Figure 3) containing vertical and horizontal reference lines vertically in the focal plane of the collimator. Verify that the objective lens is level with respect to the mounting pad. Center the reticle pattern on the target crosshairs. Failure of the reticle pattern to maintain alinement with the reference lines within one degree, or failure to meet the requirements of 3.6.9.5 shall constitute failure of this test.

4.6.9.5.5 Clarity. Focus the viewer at 25 meters distance and illuminate the scene at approximately 1 x 10^{-5} footcandles. Refocus the viewer on a target at infinity using the collimator. Failure of the reticle pattern to be sharp and clear or all numbers legible at both 25 meters and infinity focus, or failure to meet the requirements of 3.6.9.5, shall constitute failure of this test.

4.6.9.6 Battery adapter. Attach the battery adapter with known good batteries properly installed. Turn the system on with the day cover in place and observe proper illumination. Failure to illuminate, or failure to meet the requirements of 3.6.9.6 shall constitute failure of this test.

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4.6.10 High temperature storage and operation. The sight shall be subjected to Test Method 501.1, Procedure I, MIL-STD-810. Operating temperature (Step 4) shall be +52°C with the sight turned on for 3 hours. Storage temperature (Step 2) shall be +68° for 6 hours duration. The "operate", of Step 5 shall be defined as the ability to observe a green glow from the eyepiece (with the eyeguard present but detached) from the exterior of the chamber thru a window. Failure to operate per Steps 5 and 7 or pass inspection of Step 8, or failure to meet the requirements of 3.6.10, shall constitute failure of this test.

4.6.11 Low temperature storage and operation. The sight shall be subjected to Test Method 502, Procedure I, MIL-STD-810. Operating and storage temperature (Step 4) shall be -51°C for 9 hours. The "operate" of Step 5 shall be defined as the ability to turn all controls within their respective torque requirements and to observe a green glow from the eyepiece (with the eyeguard present but detached) from the exterior of the chamber thru a window. Failure to operate per Steps 5 and 7 or pass inspection of Step 8 or failure to meet the requirements of 3.6.11, shall constitute failure of this test.

4.6.12 Altitude. The sight shall be subjected to Test Method 500, Procedure I, MIL-STD-810 except that the minimum chamber pressure shall be equivalent to 10,000 feet above sea level. Operate the sight (Step 2) for 3 hours. Damage (see 3.9.1) or failure to operate per Step 2, or failure to meet the requirements of 3.6.12, shall constitute failure of this test.

4.6.13 Temperature shock. The sight shall be subjected to Test Method 503.1, Procedure I, MIL-STD-810. The maximum temperature (Steps 1 and 4) shall be +68°C and the minimum temperature (Step 2) shall be -51°C. The transition from +68°C to -51°C and from -51°C to +68°C shall include a 2 hour soak at +23°C. Steps 1,3,and 5 stabilization periods shall be a minimum of two (2) hours. Any evidence of damage (see 3.9.1) or failure to operate or to pass the inspection of Step 9, or failure to meet the requirements 3.6.13, shall constitute failure of this test.

4.6.14 Humidity. The sight shall be subjected to Test Method 507,Procedure II, MIL-STD-810 except that Step 6 shall be modified as follows. Only two continuous 48 hours cycles shall be required and at some point duringthe last 5 hours of each cycle, operate the sight and check for a green glow in the eyepiece. Failure to observe the green glow per Step 6, operate per Step 8, pass the inspection per Step 9, shall constitute failure of this test.

4.6.15 Immersion. The sight shall be subjected to Test Method 512, Procedure I, MIL-STD-810, except that the immersion period shall not be less than 30 minutes. Any evidence of damage (see 3.9.1), or failure to meet the requirements of 3.6.5 shall constitute failure of this test.

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4.6.16 High intensity shock. The sight shall be subjected to Test Method 516, Procedure IV, MIL-STD-810 in accordance with Figure 516.2-2. The shock shall be half sine pulses having a peak amplitude of 100, ± 15g's and a time duration of 4 milliseconds, ± 5 percent. Damage (see 3.9.1) or failure to pass the operational test, or failure to meet the requirements of 3.6.16 shall constitute failure of this test.

4.6.17 Interchangeability. All IPT assemblies which have passed the resolution (see 3.6.4), eyepiece diopter focus (see 3.6.9.1), and range focus (see 3.6.9.2) requirements shall be tested for interchangeability. For this test, the mount assemblies, objective lens assemblies, and eyepiece lens assemblies shall be interchanged between sight assemblies as shown in Figure 4 and the resolution (see 4.6.4), eyepiece diopter focus (see 4.6.9.1), and range focus (see 4.6.9.2) tests repeated after each test arrangement and after reassembly to the pretest arrangement. This test shall be repeated for all additional IPT assemblies until each IPT assembly has been tested once as shown in Figure 4 with another IPT assembly. Failure to meet the requirements of 3.6.17, 3.6.4, 3.6.9.1, and 3.6.9.2 after each arrangement shall constitute failure of this test.

4.7 Inspection packaging. Packaging shall be inspected to determine compliance to requirements of Section 5.

5. PACKAGING

5.1 Packaging requirements. The packaging requirements for the desired level(s) of protection shall be as specified by the acquisition activity (see 6.2.m).

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The AN/TVS-5 is intended for use at night for aimed firing of crew served weapons.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification.

b. Which of the following components are to be purchased and the quantities involved:

Right angle relay per SM-D-850181
M2 mounting bracket per SM-D-850221
M40 mounting bracket per SM-D-850231
M139, M80, tripod mounting bracket per SM-D-850241
Bag, cloth, laminated vinyl-nylon SM-D-850202-1

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Bag, cloth nylon SM-D-850202-2
Reticle M2, M85 per drawing SM-D-850490-1
Reticle M40 per drawing SM-D-850490-2
Reticle M139 per drawing SM-D-850490-6
Reticle M60 per drawing SM-D-850490-8
Batteries, BA 5567/U, BA-3058/U

c. If the image intensifier tube per MIL-I-49040 is to be purchased.

(1) To obtain the sight with Image Intensifier Tube, specify the use of drawing SM-D-850300-1.

(2) To obtain the sight without Image Intensifier Tube, specify the use of drawing SM-D-850300-2.

d. Sampling plans for Group B and Group C testing. As guidance, unless otherwise specified, the sampling plans shall conform to MIL-STD-105.

e. Qualification, if product is not qualified at time of award the contract must require qualification prior to first delivery.

f. Necessary actions by the contractor in the event of a Group C or D failure (see 4.5.4 and 4.5.6).

g. Quantity and schedule for IPT testing for all line items (see 4.4) and disposition of IPT samples.

h. The contractor is responsible to furnish sufficient M2 mounting bracket and daylight covers for the weapon shock test (see 4.6.7).

i. Prior to packing and packaging, each sight shall be examined for absence of batteries.

j. If right angle relays (see 4.6.2) are procured separately from sights, provisions shall be made in the contract to have eyepieces made available for tests.

k. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.1.2).

l. MIL-STD-810C shall be used for all environmental testing of section 4.

m. Levels of preservations and packaging (see section 5).

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6.3 Definitions. See 3.9.

6.4 Qualification. With respect to products requiring qualification, unless otherwise specified by the contract or purchase order, if a product is not qualified at time of award, the contract shall require qualification prior to first delivery. The activity responsible for approving qualified products is the CECOM, Concurrent Engineering Directorate, ATTN: AMSEL-ED-IN, Fort Monmouth, NJ 07703. Information pertaining to qualification of products may be obtained from the above activity.

6.5 Subject term keyword listing.
Image Intensification

6.6 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian:
Army - CR
Air Force - 99

Preparing activity:
Army - CR

Project 5855-0021

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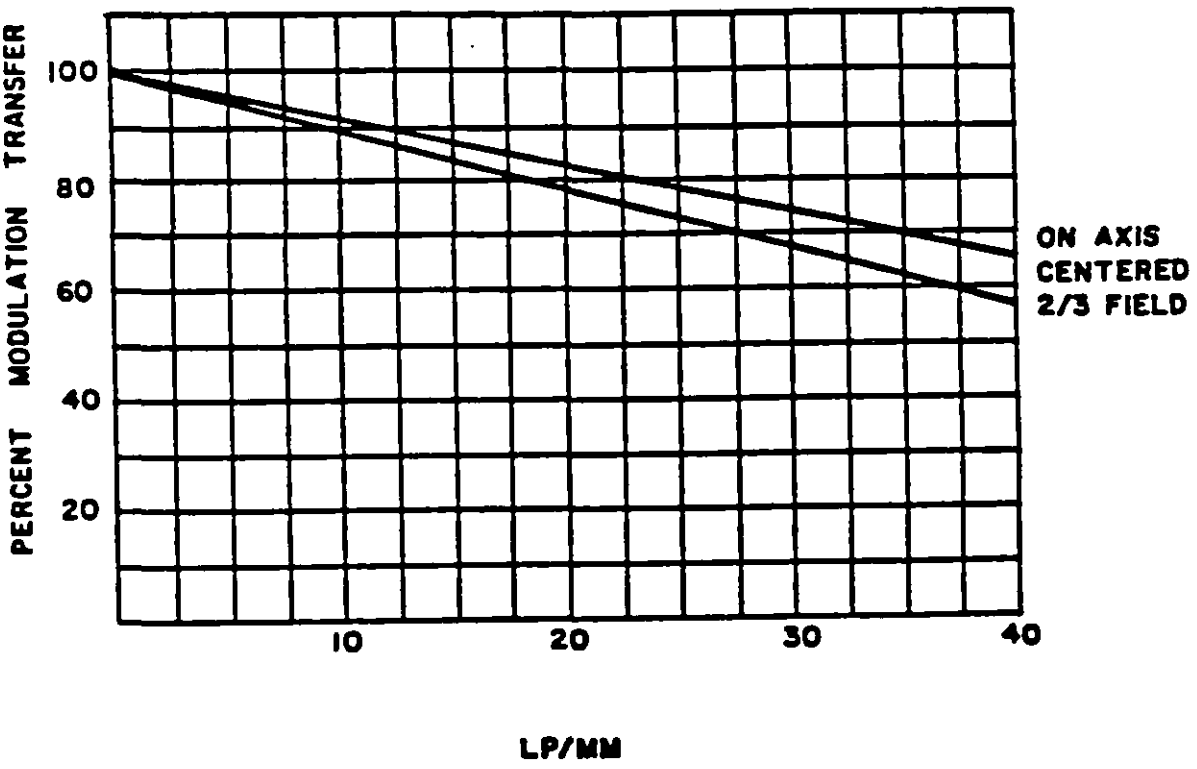


FIG. 1. RT ANGLE EYEPIECE MTF

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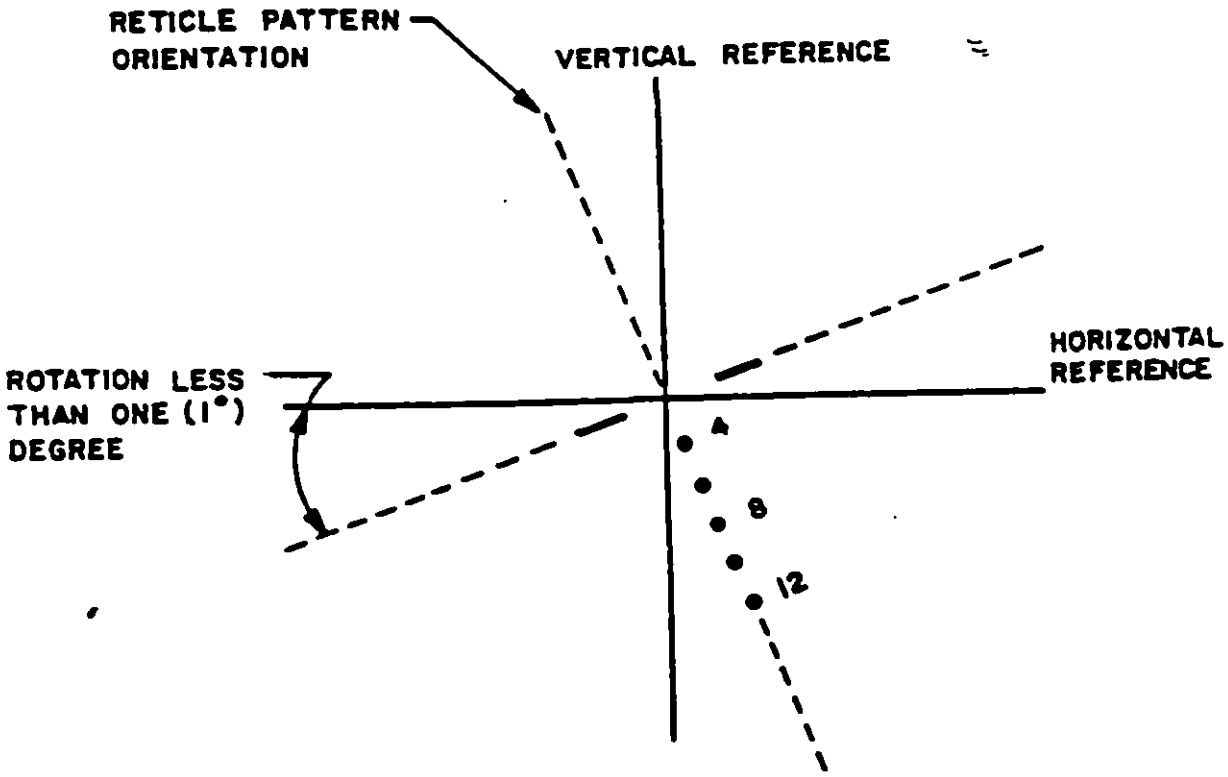
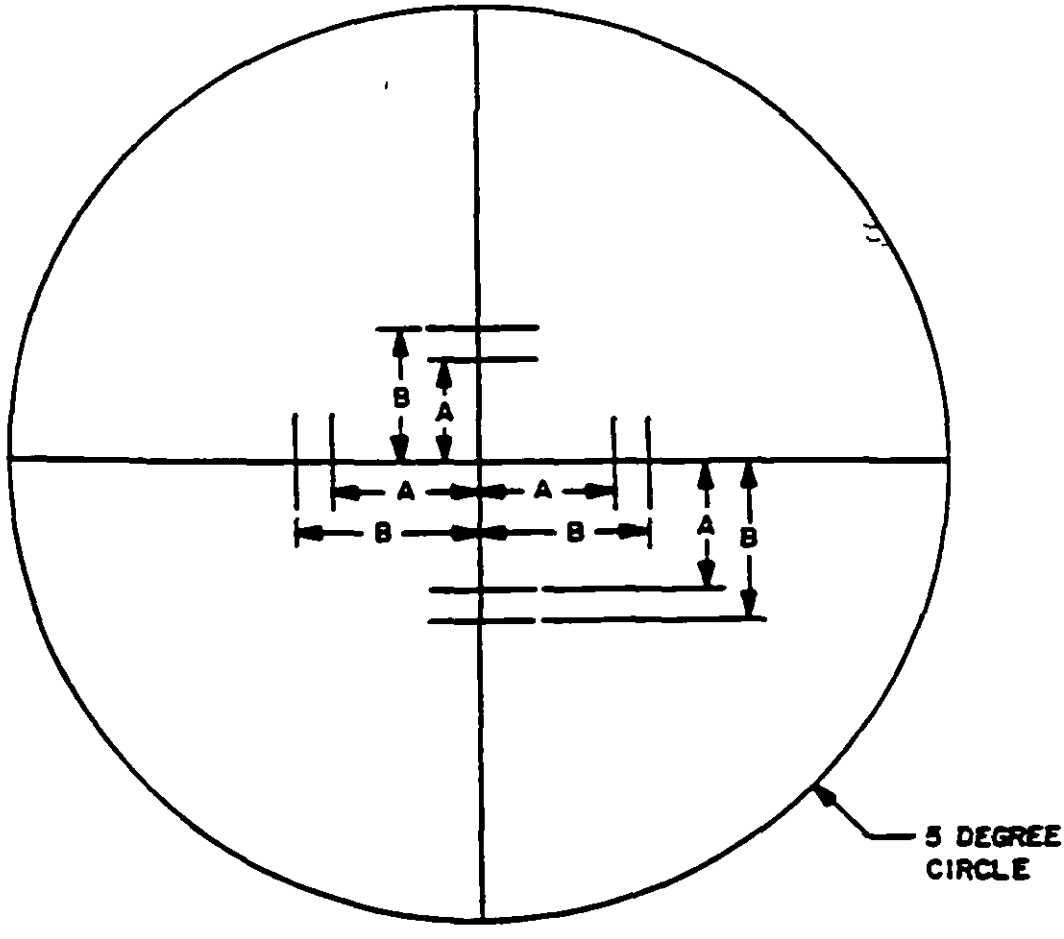


FIG. 2. RETICLE PATTERN ORIENTATION

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NOTE:

1. DIMENSION OF A EQUALS: F X 0.00425
2. DIMENSION OF B EQUALS: F X 0.00575
3. DIMENSION OF CIRCLE EQUALS: .0873 X F
4. F = FOCAL LENGTH OF COLLIMATOR

FIG. 3. TEST TARGET

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PRETEST ARRANGEMENT



TEST ARRANGEMENT 1



TEST ARRANGEMENT 2



TEST ARRANGEMENT 3



OLA - Objective Lens Assembly
MA - Mount Assembly
ELA - Eyepiece Lens Assembly

FIGURE 4. INTERCHANGEABILITY INSPECTION ARRANGEMENTS

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-N-49064D	

3. DOCUMENT TITLE
NIGHT VISION SIGHT, CREW SERVED WEAPON AN/TVS-5

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code) (1) Commercial	(2) AUTOVON
CECOM	(908) 532-5851	992-5851
c. ADDRESS (Include Zip Code) AMSEL-ED-TM Fort Monmouth, NJ 07703-5023	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340. AUTOVON 289-2340	

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